

Aniruddhan Ganesaraman

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Summary

I am a PhD student in Statistics and Operations Research (STOR) working on **causal inference**, **precision medicine**, **adaptive policy learning**, and **stochastic modeling & simulation**, driven by finding solutions to problems with meaningful real-world impact, from hospital operations to public health. I am experienced communicating technical work to varied audiences through teaching, conference talks, and UNC's Three Minute Thesis finals, and am seeking a Summer 2027 internship.

Education

Ph.D., STOR , UNC Chapel Hill ; advisors: Patrick Lopatto , Michael Kosorok	Aug 2024 – May 2029
M.S., Statistics, Analytics, and Data Science , UNC Chapel Hill ; advisor: Vidyadhar Kulkarni Thesis: <i>Data-Driven Block Replacement Scheduling</i> (arXiv)	Aug 2024 – May 2026
M.Stat. (Advanced Probability Theory), Indian Statistical Institute , Kolkata, India	Aug 2022 – May 2024
B.Sc. (Hons.), Mathematics and Computer Science , Chennai Mathematical Institute , India	Aug 2019 – May 2022

Papers & Preprints

Proactive Inpatient Bed Requests for Emergency Department Admissions <i>Manufacturing & Service Operations Management</i> ; with Qian Cheng , Nilay Tanik Argon , and Serhan Ziya . Developed data-driven policies for requesting inpatient beds earlier using patient-level predictions, reducing boarding time by 30–70% and length of stay by 6–15% in simulations. [arXiv] Presentations: INFORMS Healthcare Conference (talk, Jul 2026); INFORMS Simulation Society Research Workshop (poster, Jul 2026).	Submitted Jul 2026
Data Driven Block Replacement Scheduling <i>Operations Research</i> ; with Vidyadhar Kulkarni . Developed a data-driven method for determining the cost-minimizing preventive maintenance schedule for machine fleets from failure data, with provably sublinear regret. [arXiv] Presentation: INFORMS Annual Meeting (invited talk, Nov 2026).	Submitted Jun 2026
Heterogeneous Causal Effects of Social Determinants of Health on Cardiovascular Health <i>American Journal of Public Health</i> ; with Iheoma U. Iruka , Safoora Masoumi , Rachel M. Vaughn , and Andong Wang . Estimated, using machine learning, how these effects differ across sociodemographic groups. Presentation: American Public Health Association Annual Meeting (accepted, Nov 2026).	Submitted Aug 2026
Discussion of “Hypothesis Testing for Treatment Effect Heterogeneity via an Adaptively Augmented Debiased Estimator” <i>ASA Discoveries</i> ; with Annika Cleven , Xinyi Li , Margaret Hoch , Keshav Saxena , Matthew Egberg , Bryce Rowland , and Michael R. Kosorok .	Accepted Aug 2026

Awards & Honors

Finalist , Three Minute Thesis (3MT) 2026 , UNC Graduate School (1 of 10 university-wide)	Oct 2026
Second Place, Poster , Data Science Day 2026 , School of Data and Information Sciences , UNC <i>Proactive Inpatient Bed Requests for Emergency Department Admissions</i>	Sep 2026
First Place, Poster , YinzOR 2026 Student Conference , Carnegie Mellon INFORMS Student Chapter <i>Proactive Inpatient Bed Request Policies for Emergency Departments</i>	Aug 2026
Honorable Mention, Poster , American Causal Inference Conference 2026 <i>Estimation of a Common Local Average Treatment Effect with Multiple Instruments</i> (Tom Ten Have Poster Competition)	May 2026
Best Methodological Rigor , Center for AI and Public Health Datathon 2026 , UNC	Mar 2026
Dr. Xin Ge Graduate Student Excellence Award , UNC STOR (best first-year PhD student)	Dec 2025
Earlier: All India Rank 10, Indian Statistical Institute M.Stat. Entrance (2022); SHRIRAM Scholarship , Chennai Mathematical Institute (2019)	

Industry & Teaching Experience

Instructor of Record , STOR 155: Intro to Data Models & Inference (course website ; INDY Week)	Aug–Dec 2026
Bootcamp Instructor (Real Analysis & Linear Algebra), UNC STOR	Aug–Sep 2025
Graduate TA , UNC STOR (Graduate Probability, Stochastic Modelling)	Aug 2024 – Present
Teaching Assistant , Chennai Mathematical Institute and NPTEL (Analysis, Probability, Haskell)	Dec 2020 – Apr 2024
Data Science Intern , [24]7.ai , Bangalore: optimization algorithm for the hospitality sector	Jun–Jul 2019

Service & Leadership

STOR Liaison to Biostatistics (communication and joint activities)	Sep 2026 – Present
Organizer, STOR Graduate Student Showcase , UNC Research Week , UNC Science Expo	Oct 2026
Peer Reviewer, <i>Health Care Management Science</i> , <i>Proceedings of the American Mathematical Society</i> , <i>Involve</i>	
Head Organizer, STEMS & Tessellate , Chennai Mathematical Institute ’s math & science festival	Sep 2019 – Mar 2022

Skills

Programming: Python (pandas, NumPy, scikit-learn, PyTorch/TensorFlow), R (tidyverse, ggplot2), SQL

Tools: Git and version control, Excel, $\LaTeX$

Languages: English, Tamil, Hindi, Kannada, Bangla